



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering & Information Systems (SCORE)

Cyclesheet2-Javascript and Advanced Javascript

1. A parking garage charges a \$2.00 minimum fee to park for up to three hours. The garage charges an additional \$0.50 per hour for each hour or part thereof in excess of three hours. The maximum charge for any given 24-hour period is \$10.00. Assume that no car parks for longer than 24 hours at a time. Write a script that calculates and displays the parking charges for each customer who parked a car in this garage yesterday. You should input from the user the hours parked for each customer. The program should display the charge for the current customer and should calculate and display the running total of yesterday's receipts. The program should use the function calculate-Charges to determine the charge for each customer. Use a prompt dialog to obtain the input from the user.

2.

Write a JavaScript function that takes an array of numbers and finds the second lowest and second greatest numbers, respectively.

Sample array : [1,2,3,4,5]

Expected Output : 2,4

Write a JavaScript function that reverses a number.

Example x = 32243;

Expected Output : 34223

3. Write a JavaScript program to compute the sum of the two given integers. If the two values are the same, then return triple their sum.

4. Write a JavaScript that calculates the squares and cubes of the numbers from 0 to 10 and outputs HTML text that displays the resulting values in an HTML table format.

Number	Square	Cube
0	0	0
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
7	49	343
8	64	512
9	81	729
10	100	1000

5. Write a JavaScript program to find duplicate values in a JavaScript array.

6. Write a JavaScript program that takes an array of integers and returns false if every number is not prime. Otherwise, return true.

7.

Design a student registration form which takes student name, register number, DOB, program, email id, temporary address, permanent address, phone number. Validate the following using JavaScript:

- Mobile number should be exactly 10 digits.
- Register number should have alphabets and numbers only.

- c) Name should not exceed 30 characters and can be only alphabets.
- d) Email validation.
- e) Provide a checkbox saying "Permanent address is same as temporary address".
If checked, the value of permanent address should be added automatically from temp address. And should be in disabled mode.

8.

Write the JavaScript code to add behavior to the following page for manipulating strings.

- a) The page UI allows the user to type a phrase into a text box. The user can click the "Go!" button to display the words in that phrase in reverse order. Each word in the phrase should be inserted as a span with a class of word, inside a div with the id of words. Every other word (the first, third, fifth, etc.) should also be underlined
- b) The user can optionally specify a "filter" text by typing into a text box with the id of filter. If a non-blank filter is specified, you should exclude any words from the phrase that contain that filter text, case-insensitively. For example, if the filter text is "abc", exclude any words containing abc, ABC, aBc, etc.
- c) If any words are excluded, under the list of words you should modify the div with id of count to display text of the form, "5 word(s) filtered out". The code should work for multiple clicks of the button. On each click it should clear any previous information you injected. You may assume that words in the phrase are separated by single spaces.
- d) These screenshots show the initial state, and after phrases have been typed and "Go!" is clicked.

Sentence Reverser! Phrase: hello How ARE you? Filter: Go! you? ARE How hello	Sentence Reverser! Phrase: she sells seashells by the sea shore Filter: Go! shore sea the by seashells sells she
Sentence Reverser! Phrase: hello How ARE you? Filter: h Go! you? ARE 2 word(s) filtered out.	Sentence Reverser! Phrase: she sells seashells by the sea shore Filter: sE Go! shore the by she 3 word(s) filtered out.

9.

Write the HTML code to create the form with the following capabilities

- a) A text widget to collect the users name
- b) Four check boxes, one each for the following items
 - i. Four 100 watt light bulbs for Rs. 20=39
 - ii. Eight 100 watt light bulbs for Rs 40=20
 - iii. Four 100 watt long life light bulbs for Rs. 30=95
 - iv. Eight 100 watt long life light bulbs for Rs 70=49
- c) A collection of 3 radio buttons that are labeled as follows
 - i. Visa
 - ii. Master Card
 - iii. Discover
- d) Computes the total cost of the ordered light bulbs for the above program after adding 13.5% VAT. The program must inform the buyer of exactly what was ordered in table.

User :

Select the items : **Quantity**

☒ Four 100 watt bulbs for Rs. 20.39

☒ Eight 100watt bulbs for Rs 40.20

☐ Four 100watt long life bulbs for Rs. 30.95

☐ Eight 100watt long life bulbs for Rs 70.49

☒ VISA

☐ MASTER CARD

☐ DISCOVER

User Name:rani

You have ordered following items

Item	Qty	Price	Total
Four 100 watt bulbs for Rs. 20.39	2	20.39	40.78
Eight 100watt bulbs for Rs 40.20	1	30.95	30.95
Four 100watt long life bulbs for Rs. 30.95	0	40.2	0
Eight 100watt long life bulbs for Rs 70.49	0	70.49	0

Total bill including 13.5% tax: **81.41355** payment mode: **visa**

10.

Develop an online Railway Reservation application.

- a) It has top frame with the name “Online Railway Reservation System” in the middle, there is a logo/image in the left corner using suitable CSS styles and the left frame with the links for Ticket booking and availability checking; for each link the target should be the right frame.
- b) Create a Ticket Booking form which should have From and To in a text box, Train Names in a drop down box, Coach Number in a list box, Date in a date field, Timing in a text box, Passengers Male and female in a dropdown menu showing the count number and Submit and Reset button.
- c) Assume availability of tickets already declared in the script. Create a DOM event to display the ticket confirmation details on any HTML element when ticket is available otherwise display “tickets not available” by invoking any other DOM event.

JAVA SCRIPT EVENT HANDLING

1. Write a JavaScript program that creates a button and add a click event listener to log a message when it's clicked.
2. Write a JavaScript function that changes the background color of an element when a user moves the mouse pointer over an element.
3. Use java script to change the content of HTML tag

Figure 1 is the original HTML page. Figure 2 shows the page after clicking the button. The line above the button is changed. Requirements:

- Define a function (e.g. `changeit()`) as the event handler in the head of the html
- Register the event handler to the button (e.g. `<button ... onclick="changeit()" ...>`)
- Click the button, the content of a `<p>` is changed to "Paragraph changed"



4. Design the below calculator using Java script, html and CSS

Percentage Calculator

What is % of ?

5. Design the below web page using html, java script and CSS

Profit and Loss Calculator

Cost Price(CP) : 200

Selling Price(SP) : 300

Calculate

Profit : 100

Profit Percentage : 50.00

6. Create the below web page using java script

Enter integer x

Enter integer y

Calculate!

Output: The value of $x + y$ is 12

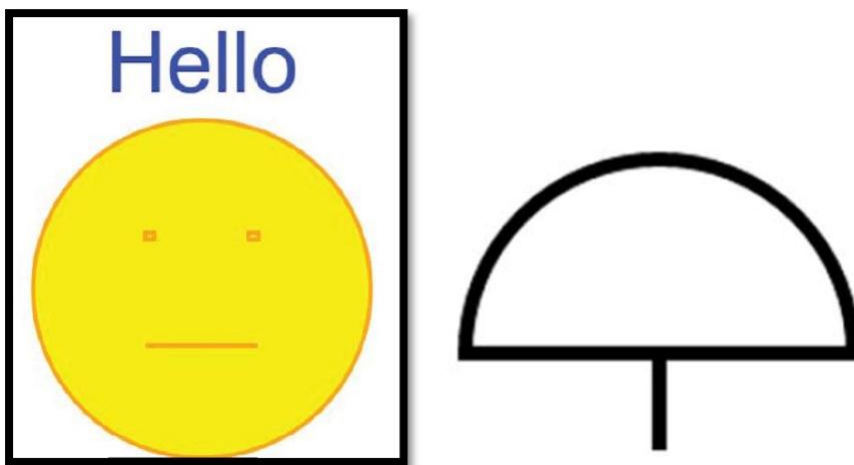
7.

Design the below page and if user clicks a box, the clicked box gets moved to front. Write a appropriate java script code to handle the click event.



8.

Write a java script to draw the below graphics on the web page



9.

Develop a JavaScript-based form validation system for a user registration form. Ensure the accurate validation of Aadhar (12 digits), PAN (format: [AAAAA1111A]), email, phone (Indian format), and password (min 8 characters, with uppercase, lowercase, numeric, and special characters). Implement real-time validation and user-friendly error messaging for a seamless user experience.